

Summable inexact-inclusion errors do not ensure convergence

Disproof of conjecture 00000008835

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The conjecture asserts that summable tolerance sequences give strong convergence for approximate inclusions. Both language versions leave this assertion unqualified. An exact proximal-point iteration is already a counterexample: the tolerances are identically zero, but the operator has no zero. This refutes the convergence conjunct; the separate, unspecified product-rate assertion is not needed.

The operator. Use the real Hilbert space $\mathbb{R}$ with its usual inner product, and the set-valued operator

$$A(x) = \{1\} \quad (x \in \mathbb{R}).$$

Its graph is $G = \{(x, 1) : x \in \mathbb{R}\}$. It is monotone, since for any two graph points

$$(x - y)(1 - 1) = 0 \geq 0.$$

It is also maximal monotone. Indeed, let a monotone graph H contain G , and take $(u, v) \in H$. Monotonicity with $(u - 1, 1)$ and $(u + 1, 1)$ gives respectively

$$v - 1 \geq 0, \quad -(v - 1) \geq 0.$$

Thus $v = 1$, so $(u, v) \in G$. No proper monotone enlargement exists.

The single-valued representative $x \mapsto 1$ is globally Lipschitz: its actual Lipschitz modulus is zero, and in particular it is 1-Lipschitz. Nevertheless

$$\text{zer } A = \emptyset.$$

No degeneracy of the proximal parameter or failure of maximality is involved.

Actual resolvent. With unit proximal parameter, the defining resolvent equation is

$$x \in y + A(y).$$

Since $A(y) = \{1\}$, this is equivalent to $x = y + 1$. It therefore has the unique solution

$$J_A(x) = (I + A)^{-1}(x) = x - 1$$

for every $x \in \mathbb{R}$.

Genuine approximate-inclusion steps. The standard additive-error proximal inclusion with tolerances is

$$u_n - u_{n+1} - e_n \in A(u_{n+1}), \quad |e_n| \leq \varepsilon_n.$$

Take an arbitrary initial point $u_0 = x$ and iterate the actual resolvent:

$$u_{n+1} = J_A(u_n), \quad e_n = 0, \quad \varepsilon_n = 0.$$

An induction gives the complete orbit

$$u_n = x - n.$$

Consequently $u_n - u_{n+1} - e_n = 1 \in A(u_{n+1})$ and $|e_n| = 0 \leq \varepsilon_n$ for every n . The tolerances are nonnegative and summable, with

$$\sum_{n=0}^{\infty} \varepsilon_n = 0.$$

This is an exact method within the approximate-inclusion framework, rather than an arbitrary sequence satisfying a postulated convergence certificate.

Failure of strong convergence. The orbit has no finite real limit for any initial point. For a direct argument, if $u_n \rightarrow a$, then also $u_{n+1} \rightarrow a$, so $u_{n+1} - u_n \rightarrow 0$. But the actual difference is constantly -1 . Uniqueness of limits would give $-1 = 0$, a contradiction. Thus summability, even zero total error, does not ensure strong convergence.

Scope and the missing hypothesis. The counterexample does not contradict a proximal-point theorem that assumes a nonempty zero set. Solvability is precisely absent from the source's unconditional convergence assertion. The operator here is everywhere defined, maximal monotone and Lipschitz, and every proximal subproblem has a unique solution. The only essential obstruction is the absence of a target zero. No assertion is made about the source's vague product of a summable rate and a Lipschitz constant.

Formal correspondence and reproduction. Lean defines the actual set-valued graph, the Hilbert inner-product monotonicity and maximality conditions, and the defining resolvent relation. It proves the unique formula $J_A(x) = x - 1$, all iterates, the complete inexact-inclusion and norm-error conditions, nonnegative summable tolerances with sum zero, and nonconvergence in the real norm topology for every start.

Run `lake build` in `lean/` using Lean 4.19.0. Public Mathlib revision:

`c44e0c8ee63ca166450922a373c7409c5d26b00b`

The project prints the relevant axiom audits.